

# Ext Groups and Ext Sheaves

Seminar Submission Note on Hartshorne III, Section 6

## Abstract

This note explains the definitions and basic properties of Ext groups and Ext sheaves in the setting of ringed spaces and noetherian schemes. The main reference is Hartshorne, *Algebraic Geometry*, Chapter III, Section 6, especially Propositions 6.2–6.9. The central theme is the distinction between the global groups

$$\mathrm{Ext}_X^i(\mathcal{F}, \mathcal{G})$$

and the local sheaves

$$\mathcal{E}xt_X^i(\mathcal{F}, \mathcal{G}).$$

The last part gives a detailed explanation of Hartshorne III, Proposition 6.9: if  $X$  is projective over a noetherian ring and  $\mathcal{F}, \mathcal{G}$  are coherent, then for  $n \gg 0$  the natural map

$$\mathrm{Ext}_X^i(\mathcal{F}, \mathcal{G}(n)) \longrightarrow \Gamma(X, \mathcal{E}xt_X^i(\mathcal{F}, \mathcal{G}(n)))$$

is an isomorphism. Conceptually, this is a consequence of the local-to-global Ext spectral sequence together with Serre vanishing. Hartshorne’s proof can also be read as an elementary induction using locally free covers and dimension shifting.

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# 1 Purpose and guiding idea

The goal is to understand two related but different objects:

$$\mathrm{Ext}_X^i(\mathcal{F}, \mathcal{G}) \quad \text{and} \quad \mathcal{E}xt_X^i(\mathcal{F}, \mathcal{G}).$$

The first is an abelian group; it is global. The second is a sheaf of  $\mathcal{O}_X$ -modules; it is local. They agree in degree zero after taking global sections,

$$\mathrm{Hom}_X(\mathcal{F}, \mathcal{G}) = \Gamma(X, \mathcal{H}om_X(\mathcal{F}, \mathcal{G})),$$

but they do not usually agree in higher degree.

A useful starting point is the identity

$$\Gamma(X, \mathcal{G}) = \mathrm{Hom}_X(\mathcal{O}_X, \mathcal{G}).$$

Taking right derived functors gives

$$\mathrm{Ext}_X^i(\mathcal{O}_X, \mathcal{G}) = H^i(X, \mathcal{G}).$$

Thus Ext groups generalize sheaf cohomology by replacing the first argument  $\mathcal{O}_X$  with an arbitrary sheaf  $\mathcal{F}$ .

By contrast, the sheaf Ext groups behave locally. In fact, under the usual noetherian and coherence hypotheses,

$$\mathcal{E}xt_X^i(\mathcal{F}, \mathcal{G})_x \cong \mathrm{Ext}_{\mathcal{O}_{X,x}}^i(\mathcal{F}_x, \mathcal{G}_x).$$

This formula shows that  $\mathcal{E}xt$  records the local commutative algebra of the stalks.

The central comparison problem is therefore:

$$\text{How is } \mathrm{Ext}_X^i(\mathcal{F}, \mathcal{G}) \text{ related to } \Gamma(X, \mathcal{E}xt_X^i(\mathcal{F}, \mathcal{G}))?$$

The answer is not equality in general. The obstruction is higher sheaf cohomology. Hartshorne III, Proposition 6.9 says that on a projective noetherian scheme this obstruction disappears after twisting the second variable sufficiently far:

$$\mathrm{Ext}_X^i(\mathcal{F}, \mathcal{G}(n)) \cong \Gamma(X, \mathcal{E}xt_X^i(\mathcal{F}, \mathcal{G}(n))) \quad (n \gg 0).$$

# 2 Notation and conventions

Throughout,  $X$  denotes a ringed space unless a stronger hypothesis is explicitly stated. The category of  $\mathcal{O}_X$ -modules is denoted by  $\mathrm{Mod}(X)$ . If  $X$  is a scheme,  $\mathcal{O}_X$  is its structure sheaf.

For sheaves  $\mathcal{F}$  and  $\mathcal{G}$  of  $\mathcal{O}_X$ -modules,

$$\mathrm{Hom}_X(\mathcal{F}, \mathcal{G})$$

denotes the abelian group of morphisms of  $\mathcal{O}_X$ -modules, while

$$\mathcal{H}om_X(\mathcal{F}, \mathcal{G})$$

denotes the sheaf Hom. Thus

$$\mathcal{H}om_X(\mathcal{F}, \mathcal{G})(U) = \mathrm{Hom}_U(\mathcal{F}|_U, \mathcal{G}|_U)$$

for open subsets  $U \subseteq X$ .

When  $X$  is equipped with an invertible sheaf  $\mathcal{O}_X(1)$ , write

$$\mathcal{G}(n) = \mathcal{G} \otimes_{\mathcal{O}_X} \mathcal{O}_X(n), \quad \mathcal{O}_X(n) = \mathcal{O}_X(1)^{\otimes n}.$$

In this note, a locally free sheaf always means a locally free sheaf of finite rank.

### 3 Definitions of Ext groups and Ext sheaves

#### 3.1 Ext groups

Fix  $\mathcal{F} \in \text{Mod}(X)$ . The functor

$$\text{Hom}_X(\mathcal{F}, -) : \text{Mod}(X) \longrightarrow \mathbf{Ab}$$

is left exact. Since  $\text{Mod}(X)$  has enough injective objects, we may define its right derived functors.

**Definition 3.1** (Ext groups). Let  $\mathcal{I}^\bullet$  be an injective resolution of  $\mathcal{G}$ :

$$0 \longrightarrow \mathcal{G} \longrightarrow \mathcal{I}^0 \longrightarrow \mathcal{I}^1 \longrightarrow \mathcal{I}^2 \longrightarrow \cdots.$$

Then

$$\text{Ext}_X^i(\mathcal{F}, \mathcal{G}) = H^i(\text{Hom}_X(\mathcal{F}, \mathcal{I}^\bullet)).$$

Equivalently,

$$\text{Ext}_X^i(\mathcal{F}, \mathcal{G}) = R^i \text{Hom}_X(\mathcal{F}, -)(\mathcal{G}).$$

The group  $\text{Ext}_X^0(\mathcal{F}, \mathcal{G})$  is just  $\text{Hom}_X(\mathcal{F}, \mathcal{G})$ .

#### 3.2 Ext sheaves

The sheaf Hom functor

$$\mathcal{H}om_X(\mathcal{F}, -) : \text{Mod}(X) \longrightarrow \text{Mod}(X)$$

is also left exact. We define its right derived functors in the same way.

**Definition 3.2** (Ext sheaves). Let  $\mathcal{I}^\bullet$  be an injective resolution of  $\mathcal{G}$ . Then

$$\mathcal{E}xt_X^i(\mathcal{F}, \mathcal{G}) = \mathcal{H}^i(\mathcal{H}om_X(\mathcal{F}, \mathcal{I}^\bullet)),$$

where  $\mathcal{H}^i$  denotes the  $i$ -th cohomology sheaf of a complex of sheaves. Equivalently,

$$\mathcal{E}xt_X^i(\mathcal{F}, \mathcal{G}) = R^i \mathcal{H}om_X(\mathcal{F}, -)(\mathcal{G}).$$

The sheaf  $\mathcal{E}xt_X^0(\mathcal{F}, \mathcal{G})$  is just  $\mathcal{H}om_X(\mathcal{F}, \mathcal{G})$ .

#### 3.3 The first warning: global Ext is not global sections of sheaf Ext

Since

$$\text{Hom}_X(\mathcal{F}, \mathcal{G}) = \Gamma(X, \mathcal{H}om_X(\mathcal{F}, \mathcal{G})),$$

one might expect that

$$\text{Ext}_X^i(\mathcal{F}, \mathcal{G}) \stackrel{?}{=} \Gamma(X, \mathcal{E}xt_X^i(\mathcal{F}, \mathcal{G})).$$

This is false in general.

The conceptual reason is that  $\text{Ext}_X^i(\mathcal{F}, \mathcal{G})$  is obtained by taking global sections of the complex  $\mathcal{H}om_X(\mathcal{F}, \mathcal{I}^\bullet)$  and then taking cohomology:

$$\text{Ext}_X^i(\mathcal{F}, \mathcal{G}) = H^i \Gamma(X, \mathcal{H}om_X(\mathcal{F}, \mathcal{I}^\bullet)).$$

On the other hand,

$$\Gamma(X, \mathcal{E}xt_X^i(\mathcal{F}, \mathcal{G})) = \Gamma(X, \mathcal{H}^i(\mathcal{H}om_X(\mathcal{F}, \mathcal{I}^\bullet))).$$

For a complex of sheaves  $C^\bullet$ , one usually does not have

$$H^i \Gamma(X, C^\bullet) \cong \Gamma(X, \mathcal{H}^i(C^\bullet)).$$

The failure is measured by sheaf cohomology.

There is, however, a natural map

$$\eta_{\mathcal{F}, \mathcal{G}}^i : \text{Ext}_X^i(\mathcal{F}, \mathcal{G}) \longrightarrow \Gamma(X, \mathcal{E}xt_X^i(\mathcal{F}, \mathcal{G})).$$

It sends a global cocycle in  $\text{Hom}_X(\mathcal{F}, \mathcal{I}^i)$  to its class in the cohomology sheaf. Proposition 6.9 says that  $\eta_{\mathcal{F}, \mathcal{G}(n)}^i$  is an isomorphism for  $n \gg 0$  in the projective noetherian coherent setting.

## 4 Basic properties of Ext groups and Ext sheaves

This section summarizes Hartshorne III, Propositions 6.2–6.8. The proofs included here are meant to explain the mechanism behind each property, not to replace the reference.

### 4.1 Restriction to open subsets: Proposition 6.2

**Proposition 4.1** (Restriction of sheaf Ext). *Let  $U \subseteq X$  be an open subset. Then there is a natural isomorphism*

$$\mathcal{E}xt_X^i(\mathcal{F}, \mathcal{G})|_U \cong \mathcal{E}xt_U^i(\mathcal{F}|_U, \mathcal{G}|_U).$$

*Idea of proof.* Choose an injective resolution  $\mathcal{G} \rightarrow \mathcal{I}^\bullet$  on  $X$ . The restriction of an injective sheaf to an open subset is injective. Therefore  $\mathcal{G}|_U \rightarrow \mathcal{I}^\bullet|_U$  is an injective resolution on  $U$ . Moreover,

$$\text{Hom}_X(\mathcal{F}, \mathcal{I}^j)|_U \cong \text{Hom}_U(\mathcal{F}|_U, \mathcal{I}^j|_U).$$

Taking cohomology sheaves gives the desired isomorphism.  $\square$

The importance of this proposition is that  $\mathcal{E}xt$  is local on  $X$ . It allows one to shrink to affine open neighborhoods when studying stalks.

### 4.2 The case $\mathcal{F} = \mathcal{O}_X$ : Proposition 6.3

**Proposition 4.2** (Ext from the structure sheaf). *For every  $\mathcal{O}_X$ -module  $\mathcal{G}$ ,*

$$\text{Ext}_X^0(\mathcal{O}_X, \mathcal{G}) \cong \mathcal{G}, \quad \text{Ext}_X^i(\mathcal{O}_X, \mathcal{G}) = 0 \quad (i > 0),$$

and

$$\text{Ext}_X^i(\mathcal{O}_X, \mathcal{G}) \cong H^i(X, \mathcal{G}).$$

*Proof.* The sheaf Hom functor satisfies

$$\text{Hom}_X(\mathcal{O}_X, \mathcal{G}) \cong \mathcal{G}.$$

Thus  $\text{Hom}_X(\mathcal{O}_X, -)$  is the identity functor on  $\text{Mod}(X)$ , hence it is exact. Its higher right derived functors vanish.

For global Ext,

$$\text{Hom}_X(\mathcal{O}_X, \mathcal{G}) = \Gamma(X, \mathcal{G}).$$

Therefore  $\text{Hom}_X(\mathcal{O}_X, -)$  is the global section functor, and its right derived functors are the sheaf cohomology groups.  $\square$

This proposition is the clearest example of the difference between global Ext and sheaf Ext. For  $i > 0$ ,

$$\mathcal{E}xt_X^i(\mathcal{O}_X, \mathcal{G}) = 0,$$

whereas

$$\text{Ext}_X^i(\mathcal{O}_X, \mathcal{G}) = H^i(X, \mathcal{G}),$$

which may be nonzero.

**Example 4.3.** If  $X$  is a smooth projective curve of genus  $g > 0$  over a field, then

$$H^1(X, \mathcal{O}_X) \neq 0.$$

Hence

$$\text{Ext}_X^1(\mathcal{O}_X, \mathcal{O}_X) \neq 0, \quad \Gamma(X, \mathcal{E}xt_X^1(\mathcal{O}_X, \mathcal{O}_X)) = 0.$$

So global Ext cannot generally be recovered from the global sections of sheaf Ext.

### 4.3 Long exact sequences in the first variable: Proposition 6.4

**Proposition 4.4** (Contravariant long exact sequences). *Given a short exact sequence*

$$0 \longrightarrow \mathcal{F}' \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}'' \longrightarrow 0,$$

*there are natural long exact sequences*

$$\begin{aligned} 0 \longrightarrow \text{Hom}_X(\mathcal{F}'', \mathcal{G}) \longrightarrow \text{Hom}_X(\mathcal{F}, \mathcal{G}) \longrightarrow \text{Hom}_X(\mathcal{F}', \mathcal{G}) \\ \longrightarrow \text{Ext}_X^1(\mathcal{F}'', \mathcal{G}) \longrightarrow \text{Ext}_X^1(\mathcal{F}, \mathcal{G}) \longrightarrow \text{Ext}_X^1(\mathcal{F}', \mathcal{G}) \longrightarrow \cdots, \end{aligned}$$

and

$$\begin{aligned} 0 \longrightarrow \mathcal{H}om_X(\mathcal{F}'', \mathcal{G}) \longrightarrow \mathcal{H}om_X(\mathcal{F}, \mathcal{G}) \longrightarrow \mathcal{H}om_X(\mathcal{F}', \mathcal{G}) \\ \longrightarrow \mathcal{E}xt_X^1(\mathcal{F}'', \mathcal{G}) \longrightarrow \mathcal{E}xt_X^1(\mathcal{F}, \mathcal{G}) \longrightarrow \mathcal{E}xt_X^1(\mathcal{F}', \mathcal{G}) \longrightarrow \cdots. \end{aligned}$$

*Idea of proof.* Let  $\mathcal{G} \rightarrow \mathcal{I}^\bullet$  be an injective resolution. For an injective sheaf  $\mathcal{I}$ , the functors

$$\text{Hom}_X(-, \mathcal{I}) \quad \text{and} \quad \mathcal{H}om_X(-, \mathcal{I})$$

are exact in the first variable. Therefore the short exact sequence in the first variable gives a short exact sequence of complexes after applying  $\text{Hom}_X(-, \mathcal{I}^\bullet)$  or  $\mathcal{H}om_X(-, \mathcal{I}^\bullet)$ . Taking cohomology yields the long exact sequences.  $\square$

The direction of the arrows reverses because Hom is contravariant in the first variable. This long exact sequence is the main tool for dimension shifting in the proof of Proposition 6.9.

### 4.4 Computing sheaf Ext by locally free resolutions: Proposition 6.5

**Proposition 4.5** (Locally free resolutions compute sheaf Ext). *Assume  $\mathcal{F}$  admits a resolution by locally free sheaves of finite rank,*

$$\cdots \longrightarrow \mathcal{L}_2 \longrightarrow \mathcal{L}_1 \longrightarrow \mathcal{L}_0 \longrightarrow \mathcal{F} \longrightarrow 0.$$

*Then*

$$\mathcal{E}xt_X^i(\mathcal{F}, \mathcal{G}) \cong h^i(\mathcal{H}om_X(\mathcal{L}_\bullet, \mathcal{G})),$$

*where  $h^i$  denotes the  $i$ -th cohomology sheaf of the displayed Hom complex.*

*Idea of proof.* If  $\mathcal{L}$  is locally free of finite rank, then

$$\mathcal{H}om_X(\mathcal{L}, -) \cong \mathcal{L}^\vee \otimes_{\mathcal{O}_X} -,$$

which is an exact functor. Therefore a locally free resolution of the first variable may be used to compute the derived sheaf Hom functors. Equivalently, this is the sheaf-theoretic analogue of computing module Ext by a projective resolution of the first variable.  $\square$

**Important warning.** A locally free resolution computes  $\mathcal{E}xt$ , but it does not directly compute global Ext. Indeed,

$$\mathrm{Hom}_X(\mathcal{L}, -) = \Gamma(X, \mathcal{L}^\vee \otimes -),$$

and the global section functor  $\Gamma(X, -)$  is not exact in general. This is why

$$\mathrm{Ext}_X^i(\mathcal{O}_X, \mathcal{G}) = H^i(X, \mathcal{G})$$

may be nonzero even though  $\mathcal{O}_X$  has a locally free resolution of length zero.

#### 4.5 Tensoring with locally free sheaves: Lemma 6.6 and Proposition 6.7

Let  $\mathcal{L}$  be a locally free sheaf of finite rank and let

$$\mathcal{L}^\vee = \mathcal{H}om_X(\mathcal{L}, \mathcal{O}_X).$$

**Lemma 4.6** (Injectives remain injective after tensoring). *If  $\mathcal{I}$  is an injective  $\mathcal{O}_X$ -module, then  $\mathcal{L} \otimes \mathcal{I}$  is injective.*

*Idea of proof.* Use the adjunction

$$\mathrm{Hom}_X(\mathcal{A}, \mathcal{L} \otimes \mathcal{I}) \cong \mathrm{Hom}_X(\mathcal{A} \otimes \mathcal{L}^\vee, \mathcal{I}).$$

The functor  $- \otimes \mathcal{L}^\vee$  is exact because  $\mathcal{L}^\vee$  is locally free of finite rank. Since  $\mathcal{I}$  is injective, the right side is an exact functor of  $\mathcal{A}$ . Hence the left side is exact in  $\mathcal{A}$ , which means  $\mathcal{L} \otimes \mathcal{I}$  is injective.  $\square$

**Proposition 4.7** (Moving a locally free factor). *There are natural isomorphisms*

$$\mathrm{Ext}_X^i(\mathcal{F} \otimes \mathcal{L}, \mathcal{G}) \cong \mathrm{Ext}_X^i(\mathcal{F}, \mathcal{L}^\vee \otimes \mathcal{G}),$$

and

$$\mathcal{E}xt_X^i(\mathcal{F} \otimes \mathcal{L}, \mathcal{G}) \cong \mathcal{E}xt_X^i(\mathcal{F}, \mathcal{L}^\vee \otimes \mathcal{G}).$$

*Idea of proof.* For degree zero this is the usual adjunction

$$\mathrm{Hom}_X(\mathcal{F} \otimes \mathcal{L}, \mathcal{G}) \cong \mathrm{Hom}_X(\mathcal{F}, \mathcal{L}^\vee \otimes \mathcal{G}),$$

and similarly for sheaf Hom. For higher degrees, take an injective resolution  $\mathcal{G} \rightarrow \mathcal{I}^\bullet$ . By the lemma,  $\mathcal{L}^\vee \otimes \mathcal{I}^\bullet$  is an injective resolution of  $\mathcal{L}^\vee \otimes \mathcal{G}$ . Applying the degree-zero adjunction term by term gives an isomorphism of complexes, hence an isomorphism on cohomology.  $\square$

A useful special case is obtained by taking  $\mathcal{F} = \mathcal{O}_X$  and replacing  $\mathcal{L}$  by a locally free sheaf  $\mathcal{E}$ :

$$\mathrm{Ext}_X^i(\mathcal{E}, \mathcal{G}) \cong \mathrm{Ext}_X^i(\mathcal{O}_X, \mathcal{E}^\vee \otimes \mathcal{G}) \cong H^i(X, \mathcal{E}^\vee \otimes \mathcal{G}).$$

Meanwhile,

$$\mathrm{Ext}_X^i(\mathcal{E}, \mathcal{G}) = 0 \quad (i > 0),$$

and

$$\mathrm{Ext}_X^0(\mathcal{E}, \mathcal{G}) = \mathcal{H}om_X(\mathcal{E}, \mathcal{G}) \cong \mathcal{E}^\vee \otimes \mathcal{G}.$$

Again, the global groups may be nonzero because they include sheaf cohomology.

## 4.6 Stalkwise computation: Proposition 6.8

**Proposition 4.8** (Stalks of sheaf Ext). *Let  $X$  be a noetherian scheme, let  $\mathcal{F}$  be coherent, and let  $\mathcal{G}$  be an  $\mathcal{O}_X$ -module. For every point  $x \in X$  there is a natural isomorphism*

$$\mathrm{Ext}_X^i(\mathcal{F}, \mathcal{G})_x \cong \mathrm{Ext}_{\mathcal{O}_{X,x}}^i(\mathcal{F}_x, \mathcal{G}_x).$$

*Idea of proof.* By restriction to open subsets, the problem is local near  $x$ . We may therefore assume that  $X$  is affine. Since  $X$  is noetherian and  $\mathcal{F}$  is coherent, one can resolve  $\mathcal{F}$  locally by finite rank free sheaves:

$$\cdots \longrightarrow \mathcal{L}_1 \longrightarrow \mathcal{L}_0 \longrightarrow \mathcal{F} \longrightarrow 0.$$

By Proposition 6.5,

$$\mathrm{Ext}_X^i(\mathcal{F}, \mathcal{G}) \cong h^i(\mathcal{H}om_X(\mathcal{L}_\bullet, \mathcal{G})).$$

Taking stalks is exact, so

$$\mathrm{Ext}_X^i(\mathcal{F}, \mathcal{G})_x \cong h^i(\mathcal{H}om_X(\mathcal{L}_\bullet, \mathcal{G})_x).$$

For locally free  $\mathcal{L}_j$ ,

$$\mathcal{H}om_X(\mathcal{L}_j, \mathcal{G})_x \cong \mathrm{Hom}_{\mathcal{O}_{X,x}}((\mathcal{L}_j)_x, \mathcal{G}_x).$$

The stalk complex  $(\mathcal{L}_\bullet)_x$  is a free resolution of  $\mathcal{F}_x$  over the local ring  $\mathcal{O}_{X,x}$ . Therefore the cohomology is precisely

$$\mathrm{Ext}_{\mathcal{O}_{X,x}}^i(\mathcal{F}_x, \mathcal{G}_x).$$

□

This proposition is one of the main reasons sheaf Ext is useful: it sheafifies the local algebraic Ext groups.

## 5 Consequences and useful interpretations

### 5.1 Local versus global information

The preceding propositions can be summarized as follows:

$$\mathrm{Ext}_X^i(\mathcal{F}, \mathcal{G}) \quad \text{is local, while} \quad \mathrm{Ext}_X^i(\mathcal{F}, \mathcal{G}) \quad \text{is global.}$$

The sheaf  $\mathrm{Ext}_X^i(\mathcal{F}, \mathcal{G})$  can be restricted to open subsets, computed on stalks, and calculated by locally free resolutions. The group  $\mathrm{Ext}_X^i(\mathcal{F}, \mathcal{G})$  also reflects global cohomology on  $X$ .



This explains the two extreme cases:

$$\mathcal{E}xt_X^i(\mathcal{O}_X, \mathcal{G}) = 0 \quad (i > 0),$$

but

$$\mathrm{Ext}_X^i(\mathcal{O}_X, \mathcal{G}) = H^i(X, \mathcal{G}).$$

Similarly, if  $\mathcal{E}$  is locally free, then

$$\mathcal{E}xt_X^i(\mathcal{E}, \mathcal{G}) = 0 \quad (i > 0),$$

but

$$\mathrm{Ext}_X^i(\mathcal{E}, \mathcal{G}) = H^i(X, \mathcal{E}^\vee \otimes \mathcal{G}).$$

## 5.2 The local-to-global Ext spectral sequence

There is a local-to-global spectral sequence

$$E_2^{p,q} = H^p(X, \mathcal{E}xt_X^q(\mathcal{F}, \mathcal{G})) \implies \mathrm{Ext}_X^{p+q}(\mathcal{F}, \mathcal{G}).$$

This spectral sequence is the conceptual framework behind the comparison between global Ext and sheaf Ext.

For total degree  $i$ , the term

$$H^0(X, \mathcal{E}xt_X^i(\mathcal{F}, \mathcal{G}))$$

is only one part of the spectral sequence. Other terms, such as

$$H^1(X, \mathcal{E}xt_X^{i-1}(\mathcal{F}, \mathcal{G})), \quad H^2(X, \mathcal{E}xt_X^{i-2}(\mathcal{F}, \mathcal{G})), \quad \dots,$$

may contribute to  $\mathrm{Ext}_X^i(\mathcal{F}, \mathcal{G})$ . Thus equality with global sections of  $\mathcal{E}xt$  should not be expected without additional vanishing.

Proposition 6.9 is exactly such a vanishing result after twisting.

## 6 Twists and Serre vanishing

From now on assume that  $X$  is projective over a noetherian ring  $A$ , and that  $\mathcal{O}_X(1)$  is a very ample invertible sheaf. Let  $\mathcal{F}$  and  $\mathcal{G}$  be coherent sheaves.

### 6.1 Serre vanishing

The form of Serre vanishing needed here is the following.

**Theorem 6.1** (Serre vanishing). *Let  $\mathcal{H}$  be a coherent sheaf on  $X$ . Then there exists an integer  $n_0$  such that for all  $n \geq n_0$ ,*

$$H^p(X, \mathcal{H}(n)) = 0 \quad \text{for all } p > 0.$$

A standard consequence is that global sections become exact after sufficiently high twisting. More precisely, if

$$0 \longrightarrow \mathcal{H}' \longrightarrow \mathcal{H} \longrightarrow \mathcal{H}'' \longrightarrow 0$$

is a short exact sequence of coherent sheaves, then for  $n \gg 0$  the sequence

$$0 \longrightarrow H^0(X, \mathcal{H}'(n)) \longrightarrow H^0(X, \mathcal{H}(n)) \longrightarrow H^0(X, \mathcal{H}''(n)) \longrightarrow 0$$

is exact. This follows from the long exact sequence of cohomology and the vanishing of  $H^1(X, \mathcal{H}'(n))$  for  $n \gg 0$ . Repeating this argument also handles finite exact sequences of coherent sheaves.

## 6.2 Compatibility of sheaf Ext with twists

For a coherent first variable, twisting the second variable commutes with sheaf Ext.

**Lemma 6.2** (Twist compatibility). *Let  $\mathcal{F}$  be coherent, let  $\mathcal{G}$  be an  $\mathcal{O}_X$ -module, and let  $\mathcal{L}$  be an invertible sheaf. Then there is a natural isomorphism*

$$\mathcal{E}xt_X^i(\mathcal{F}, \mathcal{G} \otimes \mathcal{L}) \cong \mathcal{E}xt_X^i(\mathcal{F}, \mathcal{G}) \otimes \mathcal{L}.$$

In particular,

$$\mathcal{E}xt_X^i(\mathcal{F}, \mathcal{G}(n)) \cong \mathcal{E}xt_X^i(\mathcal{F}, \mathcal{G})(n).$$

*Idea of proof.* The assertion is local on  $X$ . On an open set where  $\mathcal{L}$  is trivial, it is immediate. More explicitly, resolve  $\mathcal{F}$  locally by finite free sheaves  $\mathcal{L}_\bullet$ . Then

$$\mathcal{H}om_X(\mathcal{L}_j, \mathcal{G} \otimes \mathcal{L}) \cong \mathcal{H}om_X(\mathcal{L}_j, \mathcal{G}) \otimes \mathcal{L},$$

because  $\mathcal{L}_j$  is finite locally free. Since tensoring with an invertible sheaf is exact, the cohomology sheaves of these two complexes differ by tensoring with  $\mathcal{L}$ .  $\square$

In particular, if  $\mathcal{F}$  and  $\mathcal{G}$  are coherent on a noetherian scheme, then each  $\mathcal{E}xt_X^i(\mathcal{F}, \mathcal{G})$  is coherent, and

$$\mathcal{E}xt_X^i(\mathcal{F}, \mathcal{G}(n)) \cong \mathcal{E}xt_X^i(\mathcal{F}, \mathcal{G})(n).$$

Thus Serre vanishing applies to the sheaves  $\mathcal{E}xt_X^i(\mathcal{F}, \mathcal{G}(n))$  in the expected way.

## 7 Hartshorne III, Proposition 6.9

### 7.1 Statement

**Theorem 7.1** (Hartshorne III, Proposition 6.9). *Let  $X$  be a projective scheme over a noetherian ring  $A$ , let  $\mathcal{O}_X(1)$  be very ample, and let  $\mathcal{F}, \mathcal{G}$  be coherent sheaves on  $X$ . For each  $i \geq 0$ , the natural map*

$$\eta_{\mathcal{F}, \mathcal{G}(n)}^i : \mathcal{E}xt_X^i(\mathcal{F}, \mathcal{G}(n)) \longrightarrow \Gamma(X, \mathcal{E}xt_X^i(\mathcal{F}, \mathcal{G}(n)))$$

*is an isomorphism for all  $n \gg 0$ .*

The point is that the required lower bound for  $n$  may depend on  $i$ ,  $\mathcal{F}$ , and  $\mathcal{G}$ , but for each fixed  $i$  such a bound exists.

### 7.2 Conceptual proof by spectral sequence

The local-to-global Ext spectral sequence gives

$$E_2^{p,q}(n) = H^p(X, \mathcal{E}xt_X^q(\mathcal{F}, \mathcal{G}(n))) \implies \mathcal{E}xt_X^{p+q}(\mathcal{F}, \mathcal{G}(n)).$$

By twist compatibility,

$$\mathcal{E}xt_X^q(\mathcal{F}, \mathcal{G}(n)) \cong \mathcal{E}xt_X^q(\mathcal{F}, \mathcal{G})(n).$$

Since  $\mathcal{E}xt_X^q(\mathcal{F}, \mathcal{G})$  is coherent, Serre vanishing implies that for  $n \gg 0$ ,

$$H^p(X, \mathcal{E}xt_X^q(\mathcal{F}, \mathcal{G}(n))) = 0 \quad (p > 0).$$

Hence the spectral sequence has only the  $p = 0$  column in sufficiently high twist. It therefore degenerates, and for total degree  $i$  we get

$$\mathrm{Ext}_X^i(\mathcal{F}, \mathcal{G}(n)) \cong H^0(X, \mathcal{E}xt_X^i(\mathcal{F}, \mathcal{G}(n))).$$

This is the desired statement.

This argument explains the meaning of the theorem very clearly: twisting by  $n \gg 0$  kills all higher cohomology terms which would otherwise contribute to global Ext.

### 7.3 Hartshorne's elementary proof: base cases

Hartshorne's proof can be understood without explicitly invoking the spectral sequence. It uses Serre vanishing, long exact sequences, and induction on  $i$ .

First consider  $i = 0$ . Then

$$\mathrm{Ext}_X^0(\mathcal{F}, \mathcal{G}(n)) = \mathrm{Hom}_X(\mathcal{F}, \mathcal{G}(n))$$

and

$$\mathcal{E}xt_X^0(\mathcal{F}, \mathcal{G}(n)) = \mathcal{H}om_X(\mathcal{F}, \mathcal{G}(n)).$$

Thus

$$\mathrm{Hom}_X(\mathcal{F}, \mathcal{G}(n)) = \Gamma(X, \mathcal{H}om_X(\mathcal{F}, \mathcal{G}(n))).$$

So the result is true for  $i = 0$  for every  $n$ .

Next suppose the first variable is  $\mathcal{O}_X$ . For  $i > 0$ ,

$$\mathrm{Ext}_X^i(\mathcal{O}_X, \mathcal{G}(n)) = H^i(X, \mathcal{G}(n)),$$

while

$$\mathcal{E}xt_X^i(\mathcal{O}_X, \mathcal{G}(n)) = 0.$$

By Serre vanishing,

$$H^i(X, \mathcal{G}(n)) = 0 \quad (n \gg 0).$$

Therefore the result holds for  $\mathcal{F} = \mathcal{O}_X$ .

More generally, if  $\mathcal{E}$  is locally free of finite rank, then

$$\mathrm{Ext}_X^i(\mathcal{E}, \mathcal{G}(n)) \cong H^i(X, \mathcal{E}^\vee \otimes \mathcal{G}(n)).$$

Since  $\mathcal{E}^\vee \otimes \mathcal{G}$  is coherent, Serre vanishing gives

$$\mathrm{Ext}_X^i(\mathcal{E}, \mathcal{G}(n)) = 0 \quad (i > 0, n \gg 0).$$

On the sheaf side,

$$\mathcal{E}xt_X^i(\mathcal{E}, \mathcal{G}(n)) = 0 \quad (i > 0).$$

Thus Proposition 6.9 is true whenever the first variable is locally free.

#### 7.4 The case $i = 1$ for general coherent $\mathcal{F}$

Let  $\mathcal{F}$  be any coherent sheaf. Since  $X$  is projective over a noetherian ring, there exists a locally free sheaf  $\mathcal{E}$  of finite rank and a surjection

$$\mathcal{E} \twoheadrightarrow \mathcal{F}.$$

Let  $\mathcal{R}$  be its kernel:

$$0 \longrightarrow \mathcal{R} \longrightarrow \mathcal{E} \longrightarrow \mathcal{F} \longrightarrow 0.$$

The sheaf  $\mathcal{R}$  is coherent.

Apply  $\mathrm{Hom}_X(-, \mathcal{G}(n))$  to this short exact sequence. The long exact sequence gives

$$\begin{aligned} 0 \longrightarrow \mathrm{Hom}_X(\mathcal{F}, \mathcal{G}(n)) &\longrightarrow \mathrm{Hom}_X(\mathcal{E}, \mathcal{G}(n)) \longrightarrow \mathrm{Hom}_X(\mathcal{R}, \mathcal{G}(n)) \\ &\longrightarrow \mathrm{Ext}_X^1(\mathcal{F}, \mathcal{G}(n)) \longrightarrow \mathrm{Ext}_X^1(\mathcal{E}, \mathcal{G}(n)). \end{aligned}$$

For  $n \gg 0$ ,

$$\mathrm{Ext}_X^1(\mathcal{E}, \mathcal{G}(n)) = 0,$$

so

$$\mathrm{Ext}_X^1(\mathcal{F}, \mathcal{G}(n)) \cong \mathrm{coker}\left(\mathrm{Hom}_X(\mathcal{E}, \mathcal{G}(n)) \longrightarrow \mathrm{Hom}_X(\mathcal{R}, \mathcal{G}(n))\right).$$

On the sheaf side we obtain

$$\begin{aligned} 0 \longrightarrow \mathcal{H}om_X(\mathcal{F}, \mathcal{G}(n)) &\longrightarrow \mathcal{H}om_X(\mathcal{E}, \mathcal{G}(n)) \longrightarrow \mathcal{H}om_X(\mathcal{R}, \mathcal{G}(n)) \\ &\longrightarrow \mathcal{E}xt_X^1(\mathcal{F}, \mathcal{G}(n)) \longrightarrow \mathcal{E}xt_X^1(\mathcal{E}, \mathcal{G}(n)). \end{aligned}$$

Since  $\mathcal{E}$  is locally free,

$$\mathcal{E}xt_X^1(\mathcal{E}, \mathcal{G}(n)) = 0.$$

Thus

$$\mathcal{E}xt_X^1(\mathcal{F}, \mathcal{G}(n)) \cong \mathrm{coker}\left(\mathcal{H}om_X(\mathcal{E}, \mathcal{G}(n)) \longrightarrow \mathcal{H}om_X(\mathcal{R}, \mathcal{G}(n))\right).$$

All sheaves appearing here are coherent twists of fixed coherent sheaves. By the exactness consequence of Serre vanishing, applying  $\Gamma(X, -)$  to the sheaf sequence is exact for  $n \gg 0$ . Therefore

$$\begin{aligned} \Gamma(X, \mathcal{E}xt_X^1(\mathcal{F}, \mathcal{G}(n))) &\cong \mathrm{coker}\left(\Gamma\mathcal{H}om_X(\mathcal{E}, \mathcal{G}(n)) \longrightarrow \Gamma\mathcal{H}om_X(\mathcal{R}, \mathcal{G}(n))\right) \\ &\cong \mathrm{coker}\left(\mathrm{Hom}_X(\mathcal{E}, \mathcal{G}(n)) \longrightarrow \mathrm{Hom}_X(\mathcal{R}, \mathcal{G}(n))\right) \\ &\cong \mathrm{Ext}_X^1(\mathcal{F}, \mathcal{G}(n)). \end{aligned}$$

This proves Proposition 6.9 in degree 1.

#### 7.5 Dimension shifting: the general case

Now fix  $i \geq 2$  and use the same short exact sequence

$$0 \longrightarrow \mathcal{R} \longrightarrow \mathcal{E} \longrightarrow \mathcal{F} \longrightarrow 0.$$

The long exact sequence of global Ext contains

$$\mathrm{Ext}_X^{i-1}(\mathcal{E}, \mathcal{G}(n)) \longrightarrow \mathrm{Ext}_X^{i-1}(\mathcal{R}, \mathcal{G}(n)) \longrightarrow \mathrm{Ext}_X^i(\mathcal{F}, \mathcal{G}(n)) \longrightarrow \mathrm{Ext}_X^i(\mathcal{E}, \mathcal{G}(n)).$$

For  $n \gg 0$ , both outer terms vanish because  $\mathcal{E}$  is locally free and Serre vanishing applies:

$$\mathrm{Ext}_X^{i-1}(\mathcal{E}, \mathcal{G}(n)) = 0, \quad \mathrm{Ext}_X^i(\mathcal{E}, \mathcal{G}(n)) = 0.$$

Therefore

$$\mathrm{Ext}_X^i(\mathcal{F}, \mathcal{G}(n)) \cong \mathrm{Ext}_X^{i-1}(\mathcal{R}, \mathcal{G}(n)).$$

On the sheaf side, the long exact sequence contains

$$\mathcal{E}xt_X^{i-1}(\mathcal{E}, \mathcal{G}(n)) \longrightarrow \mathcal{E}xt_X^{i-1}(\mathcal{R}, \mathcal{G}(n)) \longrightarrow \mathcal{E}xt_X^i(\mathcal{F}, \mathcal{G}(n)) \longrightarrow \mathcal{E}xt_X^i(\mathcal{E}, \mathcal{G}(n)).$$

Since  $\mathcal{E}$  is locally free,

$$\mathcal{E}xt_X^j(\mathcal{E}, \mathcal{G}(n)) = 0 \quad (j > 0).$$

Hence

$$\mathcal{E}xt_X^i(\mathcal{F}, \mathcal{G}(n)) \cong \mathcal{E}xt_X^{i-1}(\mathcal{R}, \mathcal{G}(n)).$$

Taking global sections gives

$$\Gamma(X, \mathcal{E}xt_X^i(\mathcal{F}, \mathcal{G}(n))) \cong \Gamma(X, \mathcal{E}xt_X^{i-1}(\mathcal{R}, \mathcal{G}(n))).$$

By induction on  $i$ , Proposition 6.9 is already known for the coherent sheaf  $\mathcal{R}$  in degree  $i-1$ . Thus, for  $n \gg 0$ ,

$$\mathrm{Ext}_X^{i-1}(\mathcal{R}, \mathcal{G}(n)) \cong \Gamma(X, \mathcal{E}xt_X^{i-1}(\mathcal{R}, \mathcal{G}(n))).$$

Combining the three displayed isomorphisms gives

$$\mathrm{Ext}_X^i(\mathcal{F}, \mathcal{G}(n)) \cong \Gamma(X, \mathcal{E}xt_X^i(\mathcal{F}, \mathcal{G}(n))).$$

This completes the proof.

## 8 What Proposition 6.9 really says

The theorem does not say that global Ext is always local. It says that after sufficiently positive twisting, the global cohomological obstruction disappears.

Without twisting, the local-to-global spectral sequence has many possible contributors:

$$H^p(X, \mathcal{E}xt_X^q(\mathcal{F}, \mathcal{G})) \quad (p+q=i).$$

The term with  $p=0$  is

$$H^0(X, \mathcal{E}xt_X^i(\mathcal{F}, \mathcal{G})),$$

but terms with  $p > 0$  can also contribute. After replacing  $\mathcal{G}$  by  $\mathcal{G}(n)$ , all these higher cohomology terms vanish for  $n \gg 0$  by Serre vanishing:

$$H^p(X, \mathcal{E}xt_X^q(\mathcal{F}, \mathcal{G}(n))) = 0 \quad (p > 0).$$

Only the  $p=0$  column remains. Therefore global Ext becomes the global sections of sheaf Ext.

This is why the proposition is a natural bridge between the local algebra of  $\mathcal{E}xt$  and the global geometry of projective schemes.

## 9 Quick reference table

| Property                    | Statement                                                                                                                                                                                                         |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definitions                 | $\mathrm{Ext}_X^i(\mathcal{F}, \mathcal{G}) = R^i \mathrm{Hom}_X(\mathcal{F}, -)(\mathcal{G})$ and $\mathcal{E}xt_X^i(\mathcal{F}, \mathcal{G}) = R^i \mathcal{H}om_X(\mathcal{F}, -)(\mathcal{G})$ .             |
| Restriction                 | $\mathcal{E}xt_X^i(\mathcal{F}, \mathcal{G}) _U \cong \mathcal{E}xt_U^i(\mathcal{F} _U, \mathcal{G} _U)$ .                                                                                                        |
| Structure sheaf             | $\mathrm{Ext}_X^i(\mathcal{O}_X, \mathcal{G}) = H^i(X, \mathcal{G})$ , while $\mathcal{E}xt_X^i(\mathcal{O}_X, \mathcal{G}) = 0$ for $i > 0$ .                                                                    |
| First variable exactness    | A short exact sequence in the first variable gives long exact sequences in both global Ext and sheaf Ext, with arrows reversed.                                                                                   |
| Locally free resolution     | A locally free resolution of $\mathcal{F}$ computes $\mathcal{E}xt_X^i(\mathcal{F}, \mathcal{G})$ , but not usually $\mathrm{Ext}_X^i(\mathcal{F}, \mathcal{G})$ .                                                |
| Locally free first variable | If $\mathcal{E}$ is locally free, then $\mathcal{E}xt_X^i(\mathcal{E}, \mathcal{G}) = 0$ for $i > 0$ and $\mathrm{Ext}_X^i(\mathcal{E}, \mathcal{G}) = H^i(X, \mathcal{E}^\vee \otimes \mathcal{G})$ .            |
| Stalks                      | For $X$ noetherian and $\mathcal{F}$ coherent, $\mathcal{E}xt_X^i(\mathcal{F}, \mathcal{G})_x \cong \mathrm{Ext}_{\mathcal{O}_{X,x}}^i(\mathcal{F}_x, \mathcal{G}_x)$ .                                           |
| High twist comparison       | If $X$ is projective noetherian and $\mathcal{F}, \mathcal{G}$ are coherent, then $\mathrm{Ext}_X^i(\mathcal{F}, \mathcal{G}(n)) \cong \Gamma(X, \mathcal{E}xt_X^i(\mathcal{F}, \mathcal{G}(n)))$ for $n \gg 0$ . |

## 10 Suggested 120-minute seminar structure

| Time        | Content                                                                                                                                        |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| 0–10 min    | Motivation: Ext as a generalization of sheaf cohomology; global versus local information.                                                      |
| 10–25 min   | Definitions of $\mathrm{Ext}_X^i$ and $\mathcal{E}xt_X^i$ via injective resolutions; natural comparison map.                                   |
| 25–35 min   | Proposition 6.2: restriction to open subsets.                                                                                                  |
| 35–45 min   | Proposition 6.3: the case $\mathcal{F} = \mathcal{O}_X$ and the formula $\mathrm{Ext}_X^i(\mathcal{O}_X, \mathcal{G}) = H^i(X, \mathcal{G})$ . |
| 45–60 min   | Proposition 6.4: long exact sequences in the first variable.                                                                                   |
| 60–75 min   | Proposition 6.5: computing sheaf Ext by locally free resolutions; warning about global Ext.                                                    |
| 75–88 min   | Lemma 6.6 and Proposition 6.7: tensoring with locally free sheaves and the locally free first variable.                                        |
| 88–100 min  | Proposition 6.8: stalkwise computation and local algebra interpretation.                                                                       |
| 100–118 min | Proposition 6.9: statement, spectral-sequence intuition, and Hartshorne’s induction proof using Serre vanishing.                               |
| 118–120 min | Summary and connection to later duality theory.                                                                                                |

## 11 Checklist for understanding

A reader should be able to answer the following questions after studying the note.

1. Why does  $\mathrm{Ext}_X^i(\mathcal{O}_X, \mathcal{G})$  equal  $H^i(X, \mathcal{G})$ , while  $\mathcal{E}xt_X^i(\mathcal{O}_X, \mathcal{G})$  vanishes for  $i > 0$ ?
2. Why can a locally free resolution compute  $\mathcal{E}xt$  but not usually global Ext?
3. What does the stalk formula

$$\mathcal{E}xt_X^i(\mathcal{F}, \mathcal{G})_x \cong \mathrm{Ext}_{\mathcal{O}_{X,x}}^i(\mathcal{F}_x, \mathcal{G}_x)$$

tell us about the local nature of  $\mathcal{E}xt$ ?

4. Where does Serre vanishing enter the proof of Proposition 6.9?
5. Why does the proof of Proposition 6.9 use a locally free cover

$$0 \rightarrow \mathcal{R} \rightarrow \mathcal{E} \rightarrow \mathcal{F} \rightarrow 0?$$

6. How does dimension shifting reduce the degree  $i$  case to the degree  $i - 1$  case?

## 12 Conclusion

Ext groups and Ext sheaves are two versions of the same derived construction. The group  $\mathrm{Ext}_X^i(\mathcal{F}, \mathcal{G})$  is global and includes sheaf cohomology, while the sheaf  $\mathcal{E}xt_X^i(\mathcal{F}, \mathcal{G})$  is local and can be read on stalks as a module Ext group over the local ring. Their difference is exactly the kind of difference that sheaf cohomology measures.

Hartshorne III, Proposition 6.9 shows that on a projective noetherian scheme, coherent sheaves become well behaved after sufficiently positive twisting. Serre vanishing kills the higher cohomology terms in the local-to-global Ext spectral sequence, so the global Ext group is recovered from the global sections of the sheaf Ext:

$$\mathrm{Ext}_X^i(\mathcal{F}, \mathcal{G}(n)) \cong \Gamma(X, \mathcal{E}xt_X^i(\mathcal{F}, \mathcal{G}(n))) \quad (n \gg 0).$$

This result prepares the ground for later applications, especially duality theory, where Ext groups become the natural objects appearing in global duality statements.

## References

- [1] R. Hartshorne, *Algebraic Geometry*, Graduate Texts in Mathematics 52, Springer, 1977. See Chapter III, Section 6.